KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)

Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A++" Grade



COURSE: DATA COMMUICATION NETWORKS

Problem-Theme: Network Performance Optimization

Class: III C

Solution Title: Intelligent Network Performance Optimization Using ML

Team No: 5

CAUSES

- Rapid increase in heterogeneous traffic (VoIP, HTTP, FTP)
- Heavy FTP traffic consumes most of the bandwidth
- No traffic prioritization in traditional networks
- Congestion at bottleneck routers
- High delay, packet loss, and jitter affecting real-time applications

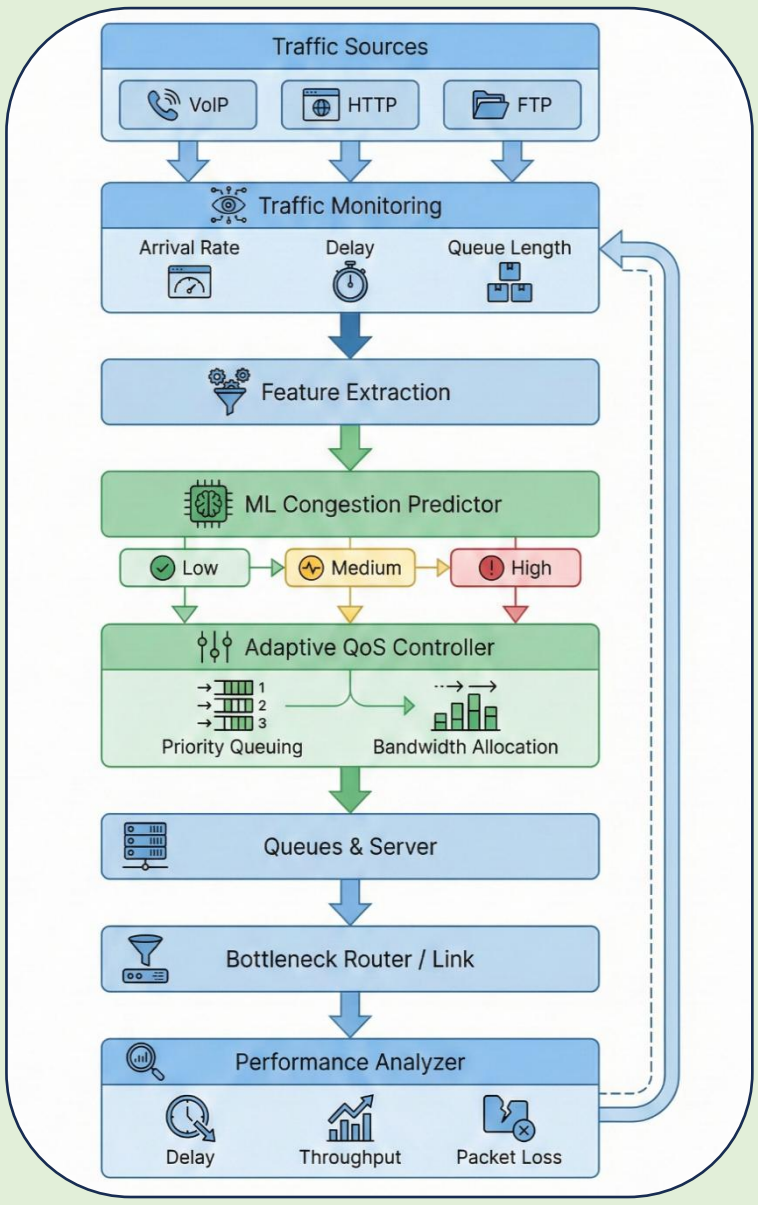
SOLUTIONS

- Bandwidth Upgradation
- Quality of Service (QoS)
- Traffic Shaping & Policing
- Congestion Control Mechanisms
- Load Balancing
- Network Optimization Algorithms
- Software Defined Networking

METHODOLGY

- Intelligent QoS-based approach where network traffic is prioritized dynamically using Machine Learning to handle congestion during peak hours.
- Monitor network traffic (VoIP, HTTP, FTP) in real time
- Extract key parameters (traffic rate, delay, queue length)
- Using Machine Learning to predict congestion level
- Apply QoS-based traffic prioritization dynamically
- Allocate higher priority and bandwidth to real-time traffic
- Reduce delay and packet loss during peak hours

Block Diagram



Team Members:

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LITERATURE SURVEY

1. **"A. Al-Zubaidi, M. A. Abbas, and H. A. Naman", Network traffic classification to improve quality of service (QoS)**
   6th International Conference for Physics and Advanced Computation Sciences (ICPAS 2024).
   Link: https://www.researchgate.net/publication/390514669_Network_traffic_classification_to_improve_quality_of_service_QoS
2. **"R. H. Serag, A. M. Abd El-Latif, and S. A. El-Sayed", Software Defined Network Traffic Classification for QoS Optimization Using Machine Learning**
   Journal of Network and Systems Management, Springer, 2025.
   Link: https://link.springer.com/article/10.1007/s10922-025-09911-6?utm_source
3. **"M. F. Osman, A. R. Hassan, and S. A. Ibrahim", Optimizing Network Performance and Quality of Service with AI**
   International Journal of Future Electronics and Technology Research (IJFETR), 2024.
   Link: https://frontiersrj.com/journals/ijfetr/sites/default/files/IJFETR-2024-0041.pdf
4. **"Y. C. Chen, C. H. Lin, and Y. S. Lin", A Machine Learning Based Approach to QoS Metrics Prediction**
   Journal of Communications, Vol. 34, No. 3, 2022.
   Link: https://csroc.org.tw/journal/JOC34-3/JOC3403-15.pdf
5. **"M. R. Hossain, S. Islam, and M. S. Rahman", Predicting Quality of Service on Cellular Networks Using AI**
   Journal of Electrical Engineering and Informatics, 2023.
   Link: https://journalcenter.org/index.php/jeei/article/download/3901/3062/14125
6. **"A. Rostami, M. Gidlund, and T. Lennvall", Machine-Learning Assisted Provisioning of Time-Varying Traffic**
   Chalmers University of Technology, Research Publication, 2022.
   Link: https://research.chalmers.se/publication/542493/file/542493_Fulltext.pdf
7. **"N. Kumar and A. Ahmad", ML-Based QoS and Traffic-Aware Prediction-Assisted Network Slicing**
   International Journal of Communication Networks & Distributed Systems, 2022.
   Link: https://dl.acm.org/doi/10.1504/IJCNDS.2022.120298
8. **"G. Wassie, A. Schoesser, and H. D. Schotten", Traffic Prediction in SDN for Explainable QoS Using Deep Models**
   Scientific Reports (Nature), 2023.
   Link: https://www.nature.com/articles/s41598-023-46471-8

Literature summary:

- ➔ Peak-hour network congestion increases delay, packet loss, and degrades QoS, especially for real-time applications.
- ➔ Traditional solutions based on bandwidth upgradation and static routing are costly and not scalable.
- ➔ QoS techniques such as traffic classification, priority queuing, and bandwidth allocation improve real-time performance but are reactive.
- ➔ Recent research shows that Machine Learning enables predictive congestion control and improves network decision-making.
- ➔ ML-assisted QoS consistently outperforms static QoS, achieving lower delay and higher throughput.

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